

industries such as health, these now form critical parts of life-saving mechanisms and emergency systems. Any failure in such devices in a critical juncture of life and death could lead to the loss of life of a patient.

With remote sensors and monitoring materialising as the core of the IoT, data control and ownership in business organisations will become more sensitive. This will drive compliance issues to become pertinent across every industry vertical. Unique compliance frameworks developed and customised according to the unique requirements of businesses will emerge.

For adequate data protection, development of new security frameworks will gain an evolutionary trend, as these will have to evolve consecutively to deal with new and evolving security threats. In fact, the evolutionary nature of the IoT is set to complicate the industry. With a huge number of players in the IoT, enterprises will have to follow multiple standards and protocols along with a large number of compliances.

The benefits

Despite its complexities, the IoT has been successful in creating hype primarily because of the advantages it offers. Cost saving, improved process operations, higher productivity, obtaining real-time information, forming massive networks to overcome geographical boundaries and innovations in businesses are the primary advantages offered by this new technology.

The IoT brings down the operational cost of a business substantially, as it sidelines the need for expensive infrastructure by encouraging an extended utilisation of the cloud. Businesses are becoming more automated, resulting in efficient and agile process structures. The IoT eradicates possible human

errors as human intervention becomes less relevant in business.

By improving these aspects, businesses can ensure higher productivity and reap better profits out of their operations. The IoT has facilitated business expansion across large geographical locations and offers more pervasive services or products to customers. As it improves supply chain management and facilitates real-time tracking, enterprises gain radical advantages to manage their businesses through the IoT.

The way forward

To harness the potentials of the IoT optimally, threads and complexities in this technology must be addressed intuitively. Threat-centric and automated approaches to security must be leveraged to gain maximum advantage out of smart IoT networks.

Operational technology (OT) of used devices must be integrated to IT platforms within businesses after intensive scrutiny and research. As OT layers in these devices are the most vulnerable entry points for malicious agents, more intensive technology must be developed aimed at increasing its integrity with Internet security protocols.

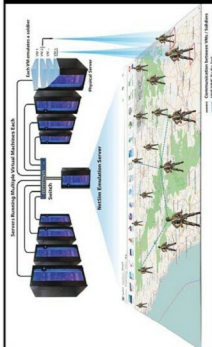
Increased operational efficiency offered by the IoT could offer a win-win situation to businesses as well as consumers, while improving their quality of life. Effectual intelligence can be generated using combined information from smart devices.

However, the emergent network structure is also advocating sheer vulnerabilities to the whole infrastructure. As sensitive information critical to businesses as well as personal lives of users will be exposed in the network or in the cloud, without a strong and properly maintained security infrastructure, fear of data leakage to malicious entities will always exist. **EFY**

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